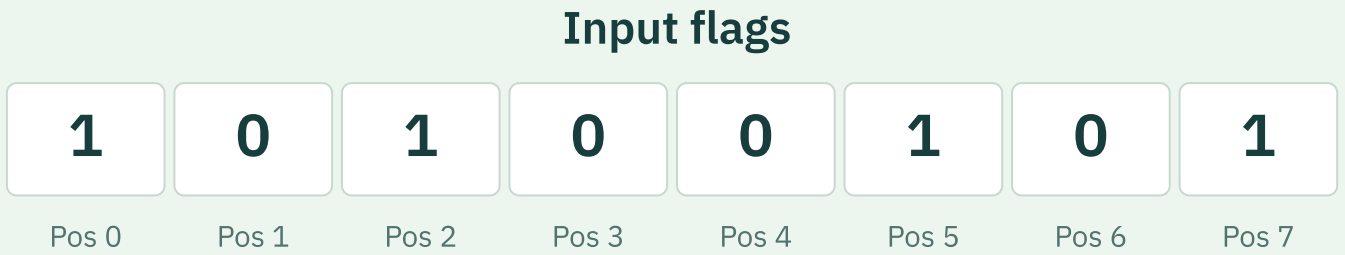
1 / 6

Each token needs its own prefix.

A flag selects a token; a prefix counts selected inputs.

> Snapshot-ordered integer scan · one block · dependency stages, not GPU cycles



Inclusive: include the current flag

Exclusive: count only earlier flags

One prefix total for every input position

Integer flags · one block · no GPU timing

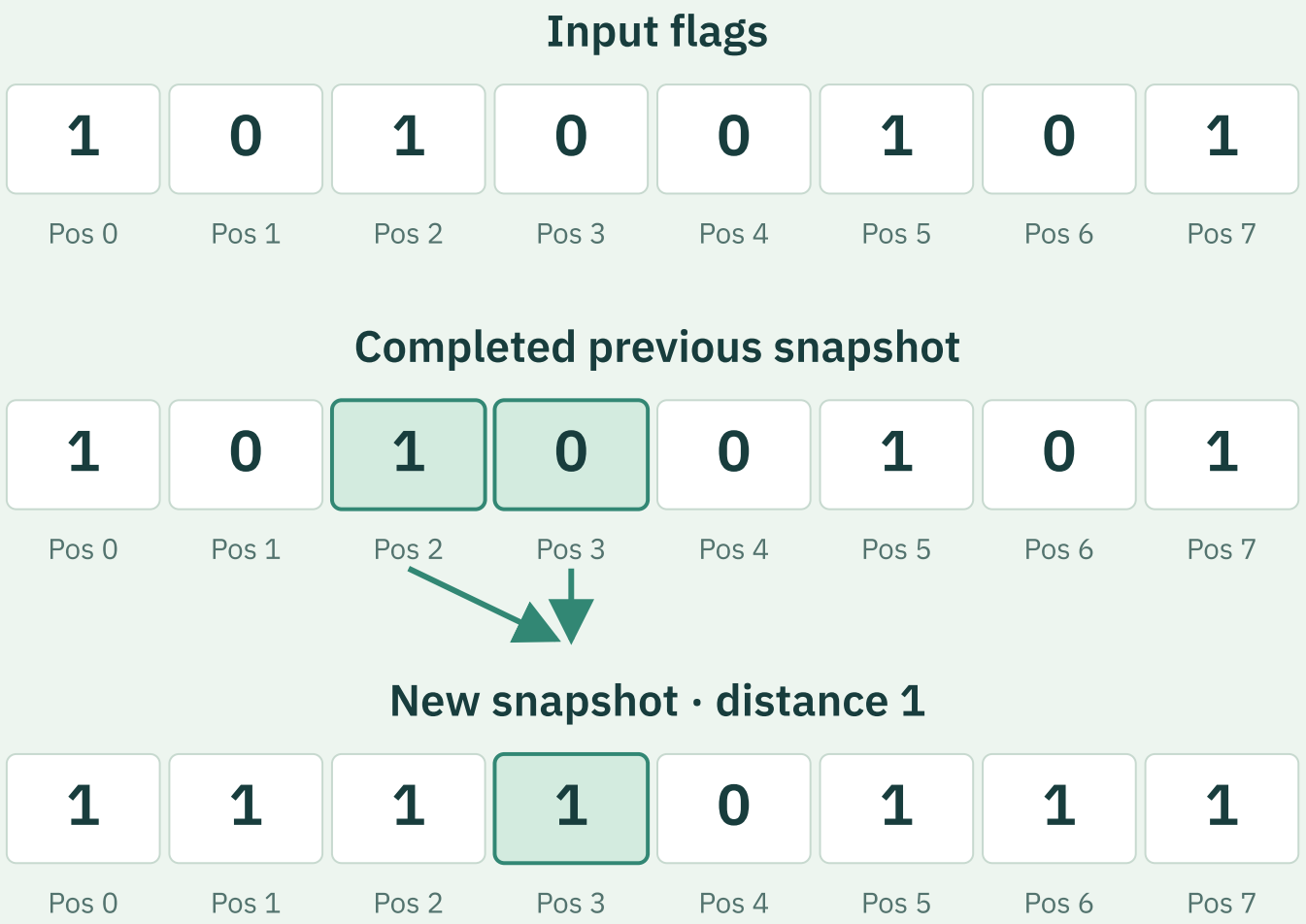
Copyright © 2026 Christopher Snow

2 / 6

Read a completed previous snapshot.

At distance 1, combine each value with its earlier neighbour.

> Snapshot-ordered integer scan · one block · dependency stages, not GPU cycles



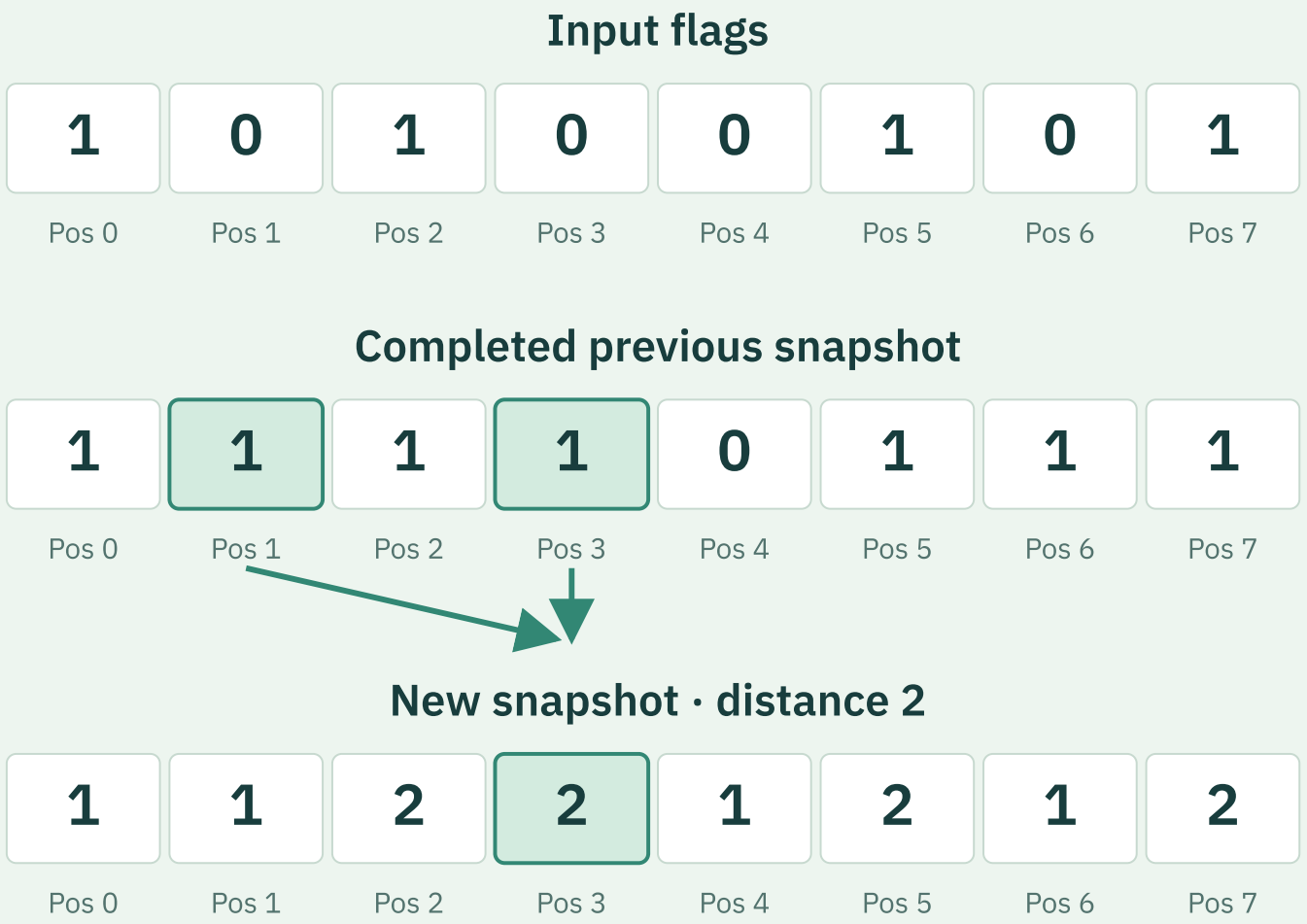
Copyright © 2026 Christopher Snow

3 / 6

Double the covered prefix.

At distance 2, combine completed two-input groups.

> Snapshot-ordered integer scan · one block · dependency stages, not GPU cycles



Copyright © 2026 Christopher Snow

4 / 6

An offset counts earlier survivors.

Shift inclusive prefixes right and insert zero.

> Snapshot-ordered integer scan · one block · dependency stages, not GPU cycles



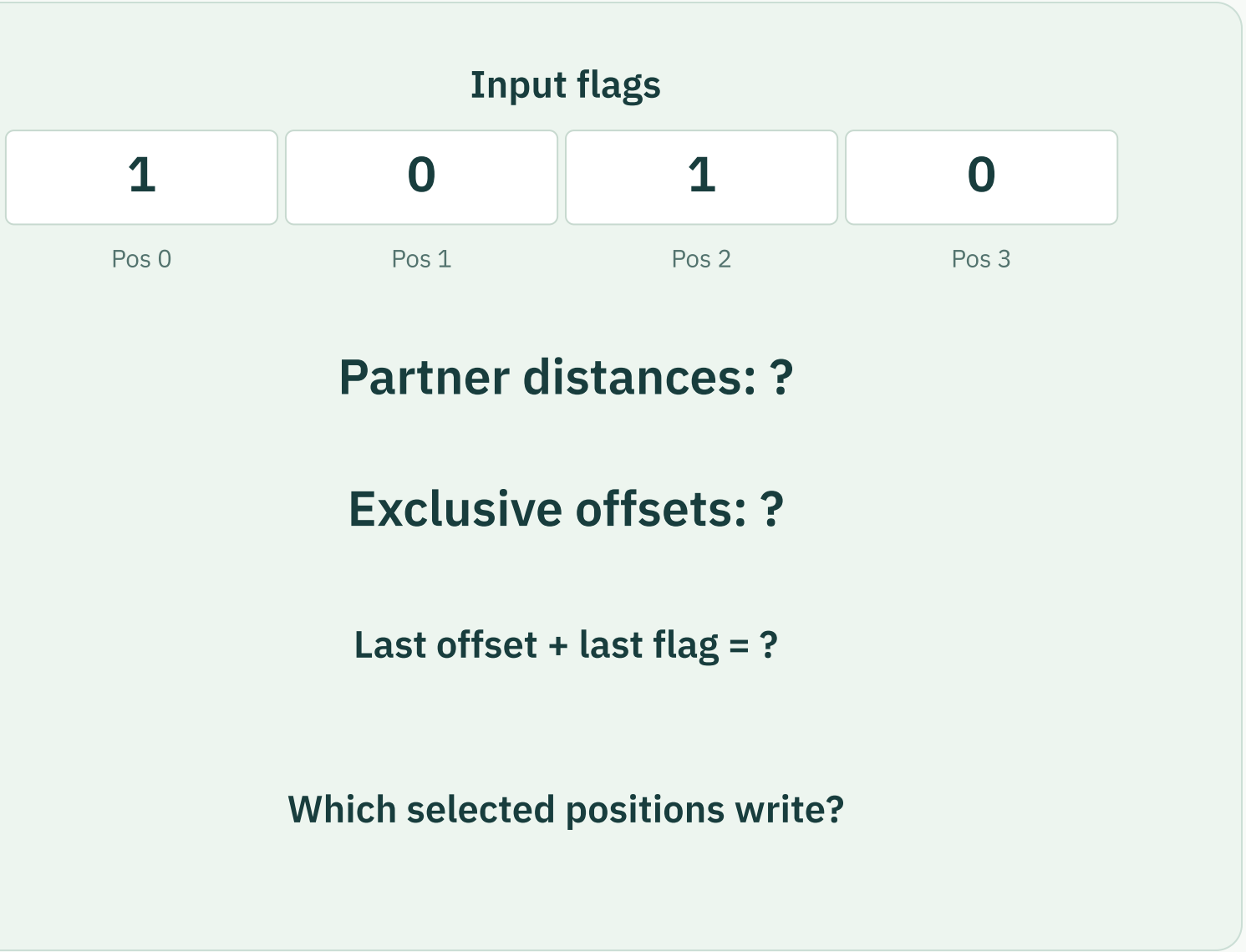
Copyright © 2026 Christopher Snow

5 / 6

Pause: keep only the first four flags.

Predict the stages and the exclusive output count.

> Snapshot-ordered integer scan · one block · dependency stages, not GPU cycles



Copyright © 2026 Christopher Snow

6 / 6

Two stages give the four-input prefixes.

Only selected positions use their offsets for writes.

> Snapshot-ordered integer scan · one block · dependency stages, not GPU cycles



Copyright © 2026 Christopher Snow